

# Conjecture 5

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id 1602.05184\_\_00 — generated 2026-09-17

**UNREFEREED MODEL OUTPUT.** This document was selected solely because GPT-6 Astra labelled its own result would\_publish: true and returned proved or disproved. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

## Summary and provenance

|                        |                                                                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Catalog id             | 1602.05184__00                                                                                                                                       |
| Catalog entry          | <a href="#">Conjecture 5</a>                                                                                                                         |
| Source corpus          | arxiv                                                                                                                                                |
| Campaign leg           | selected second attempts (attacks_retry)                                                                                                             |
| Source paper / entry   | On the difference between the Szeged and Wiener index                                                                                                |
| Model verdict          | <b>proved</b> (confidence: high)                                                                                                                     |
| Model's one-line claim | Every 2-connected graph outside the three exceptional families satisfies $\eta(G) \geq \min\{2n, 3n-10\}$ , proving the conjecture for $n \geq 10$ . |
| Model                  | gpt-6-astra, reasoning effort max, mode=pro, flex service tier                                                                                       |
| Original artifact      | attacks_retry/1602.05184__00/output.md                                                                                                               |
| Independent review     | <b>None</b>                                                                                                                                          |
| Model's caveats        | The proof is self-contained; novelty and current literature status have not been independently checked.                                              |

## Problem statement

No extracted statement was stored in the artifact metadata.

## Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

### 1. Statement

All graphs below are finite and simple. Write  $K_n^t$  for a clique of order  $n - 1$  together with an additional vertex adjacent to exactly  $t$  clique vertices.

I prove the following stronger statement.

**Theorem**

Let  $G$  be a 2-connected graph of order  $n \geq 4$ , with

$$G \not\cong K_n, K_n^2, K_n^{n-2}.$$

Then

$$\boxed{\eta(G) \geq \min\{2n, 3n - 10\}.} \quad (1)$$

Consequently, if  $n \geq 10$ , then  $\eta(G) \geq 2n$ .

The universal-vertex argument in the supplied attempt is correct; a verification is included below. The additional ingredient is a reduction for graphs without universal vertices. It is important that this reduction produces a dominated vertex **whose deletion remains 2-connected**.

**2. Good edges and dominated-vertex deletion**

For an edge  $uv$ , let  $n_{uv}(u)$  denote the number of vertices closer to  $u$  than to  $v$ . Thus

$$Sz(G) = \sum_{uv \in E(G)} n_{uv}(u)n_{uv}(v), \quad W(G) = \sum_{\{a,b\} \subseteq V(G)} d(a,b).$$

An edge  $uv$  is **good** for  $\{a, b\}$  if, after possibly exchanging  $u, v$ ,

$$d(a, u) < d(a, v), \quad d(b, v) < d(b, u).$$

Let  $g_G(a, b)$  be the number of good edges, and put

$$D_G(a, b) = g_G(a, b) - d_G(a, b), \quad c_G(a) = \sum_{b \neq a} D_G(a, b).$$

Every edge of every shortest  $a$ - $b$  path is good for  $\{a, b\}$ . In particular,

$$D_G(a, b) \geq 0.$$

Double counting gives

$$\boxed{\eta(G) = \sum_{\{a,b\}} D_G(a, b) = \frac{1}{2} \sum_a c_G(a).} \quad (2)$$

We shall repeatedly use the following elementary deletion facts.

**Lemma 2.1**

Suppose that  $G$  is 2-connected,  $|V(G)| \geq 4$ , and

$$N_G[u] \subseteq N_G[v] \quad (u \neq v).$$

Put  $H = G - u$ .

1.  $H$  is isometric in  $G$ .
2. The only possible cut vertex of  $H$  is  $v$ .
3. If  $q_G(u)$  is the number of incidences

$$(\{a, b\}, e)$$

where  $a, b \in V(H)$  and  $e$  is an edge incident with  $u$  that is good for  $\{a, b\}$ , then

$$\eta(G) - \eta(H) = c_G(u) + q_G(u). \quad (3)$$

4. If  $u$  is simplicial, then  $q_G(u) = 0$ . If  $u$  is not simplicial, then  $q_G(u) \geq 2$ .

**Proof** Any portion  $x - u - y$  of a path between vertices of  $H$  can be replaced by a walk through  $v$  of length at most two. This proves that deleting  $u$  preserves all distances between the remaining vertices.

Suppose  $w \neq v$  were a cut vertex of  $H$ . All vertices of  $N_G(u) \setminus \{w\}$  lie in the component of  $H - w$  containing  $v$ , since they are adjacent to  $v$ . Adding  $u$  therefore cannot connect the components of  $H - w$ , contradicting the connectivity of  $G - w$ . This proves the second assertion.

Because distances in  $H$  are unchanged, an edge of  $H$  is good for an old pair in  $G$  exactly when it is good for that pair in  $H$ . The additional contributions are precisely the pairs containing  $u$ , and the incidences counted by  $q_G(u)$ . Hence (3).

If  $u$  is simplicial, then for every  $x \in N(u)$  and every  $a \neq u$ ,

$$d(a, x) \leq d(a, u).$$

Indeed, take the last neighbor of  $u$  on a shortest  $a$ - $u$  path and use that  $N(u)$  is a clique. Thus no vertex other than  $u$  is closer to  $u$  than to  $x$ , and  $ux$  cannot be good for a pair avoiding  $u$ .

Finally, if  $u$  is not simplicial, choose nonadjacent  $x, y \in N(u)$ . Both  $xu$  and  $uy$  are good for  $\{x, y\}$ , giving  $q_G(u) \geq 2$ .  $\square$

We also use the standard consequence that deleting a simplicial vertex from a 2-connected graph of order at least four preserves 2-connectivity. For completeness, if deletion created a cut vertex  $w$ , all remaining neighbors of the simplicial vertex would lie in one component after deleting  $w$ , so restoring it could not reconnect the graph.

### 3. The structure of a vertex with contribution at most three

The next lemma is the main structural step.

#### Lemma 3.1 — Low-contribution structure

Let  $G$  be 2-connected, and let  $a$  be a non-universal vertex satisfying  $c_G(a) \leq 3$ . Put  $A = N_G(a)$ .

Every vertex has distance at most two from  $a$ , and one of the following descriptions holds.

**Type I** There are adjacent vertices  $p, q \in A$  and a vertex  $z \notin N[a]$  such that

$$V(G) \setminus N[a] = \{z\} \dot{\cup} X, \quad N(z) \cap A = \{p, q\}.$$

Either  $X = \emptyset$ , or:

- $G[X]$  is connected;
- every vertex of  $X$  has neighborhood in  $A$  exactly  $\{p\}$ ;
- there is exactly one edge between  $z$  and  $X$ .

**Type II** There are distinct  $p, q \in A$  and nonempty sets  $X, Y$  such that

$$V(G) \setminus N[a] = X \dot{\cup} Y,$$

where:

- $G[X]$  and  $G[Y]$  are connected;
- every vertex of  $X$  has neighborhood in  $A$  exactly  $\{p\}$ ;
- every vertex of  $Y$  has neighborhood in  $A$  exactly  $\{q\}$ ;
- there is exactly one edge  $xy$  between  $X$  and  $Y$ .

The edges inside  $A$ , and inside the indicated connected sets, are otherwise unrestricted.

Moreover, in any 2-connected graph,

$$a \text{ non-universal} \implies c_G(a) \geq 2. \quad (4)$$

**Proof** Use distance layers  $L_i$  from  $a$ . A neighbor of a vertex in the preceding layer will be called a **predecessor**.

Let  $M$  be the set of vertices having more than one shortest path from  $a$ . If  $b \in M$ , the union of two distinct shortest  $a$ - $b$  paths contains at least  $d(a, b) + 2$  edges, all good for  $\{a, b\}$ . Hence

$$D_G(a, b) \geq 2. \quad (5)$$

It follows that  $|M| \leq 1$ .

We first record another useful observation. Suppose  $x, y \in L_i$  have unique shortest paths from  $a$ , their predecessors are different, and  $xy \in E(G)$ . If  $p(y)$  denotes the predecessor of  $y$ , then

$$d(x, y) = 1, \quad d(x, p(y)) = 2.$$

Thus  $p(y)y$  is good for  $\{a, x\}$ , and it does not belong to the unique shortest  $a$ - $x$  path. The analogous assertion holds with  $x, y$  exchanged.

For a fixed  $x$ , different such neighbors  $y$  give different extra good edges.

**Case 1:**  $M = \emptyset$

The edges between consecutive layers form a rooted tree  $T$ : every non-root vertex has exactly one predecessor.

Let  $F$  be the set of same-layer edges whose endpoints have different predecessors. The preceding observation gives

$$c_G(a) \geq 2|F|. \quad (6)$$

Thus  $|F| \leq 1$ .

If  $F = \emptyset$ , all non-tree edges join siblings. In such a graph, the proper descendants of any non-root tree vertex have no neighbors outside that subtree except its root. Consequently, a vertex at distance at least two from  $a$  would force a cut vertex. Since  $G$  is 2-connected,  $a$  would be universal, contrary to assumption.

Therefore  $F = \{xy\}$ , say  $x, y \in L_i$ , with distinct predecessors  $p, q$ .

We claim that  $p, q$  have the same predecessor. Otherwise  $p, q$  are nonadjacent, since  $pq$  would be a second edge of  $F$ . They also have no common neighbor:

- a common neighbor in  $L_{i-2}$  contradicts their distinct predecessors;
- a common neighbor in  $L_{i-1}$  would produce another edge of  $F$ ;
- a common neighbor in  $L_i$  would have two shortest paths from  $a$ .

Hence

$$d(p, q) = 3$$

via  $p - x - y - q$ , while  $d(p, y) = d(q, x) = 2$ . Therefore  $qy$  is an extra good edge for  $\{a, p\}$ , and  $px$  is an extra good edge for  $\{a, q\}$ . Together with the extra contributions at  $x$  and  $y$ , this gives  $c_G(a) \geq 4$ , a contradiction.

Let  $w$  be the common predecessor of  $p, q$ . Apart from sibling edges, the only non-tree edge is  $xy$ , which lies entirely among proper descendants of  $w$ . If  $w \neq a$ , those descendants are separated from  $a$  by  $w$ , contradicting 2-connectivity. Thus  $w = a$ , and  $i = 2$ .

There can be no vertex in  $L_3$  or a later layer. Indeed, the proper descendants of its ancestor in  $L_2$  would have no exit except through that ancestor: the sole exceptional edge  $xy$  has both endpoints in  $L_2$ .

Every vertex of  $L_2$  therefore has a unique neighbor in  $A$ . The only edge between different predecessor classes is  $xy$ . A class other than those of  $p, q$ , or a component of the  $p$ -class not containing  $x$ , would be separated by its predecessor. The analogous statement holds for  $q$ . This gives Type II.

**Case 2:**  $M = \{z\}$

The vertex  $z$  has at least two predecessors; otherwise its unique predecessor would also belong to  $M$ . It has no neighbor in the next layer, since such a neighbor would also belong to  $M$ .

If  $z$  had  $r \geq 3$  predecessors, the last two layers of its shortest paths would contain at least  $2r$  good edges, only two of which belong to one fixed shortest path. Hence

$$D_G(a, z) \geq 2r - 2 \geq 4,$$

which is impossible. Thus  $z$  has exactly two predecessors, say  $p, q$ .

If  $pq \notin E(G)$ , then  $qz$  is an extra good edge for  $\{a, p\}$ , and  $pz$  is one for  $\{a, q\}$ . Together with  $D_G(a, z) \geq 2$ , this again gives  $c_G(a) \geq 4$ . Hence  $pq \in E(G)$ .

There cannot be a same-layer edge between two uniquely reached vertices with different predecessors: it would contribute at least one at each endpoint in addition to the contribution of at least two at  $z$ . In particular,  $p, q$  have the same predecessor, say  $w$ .

Consider a same-layer neighbor  $t$  of  $z$ . Its unique predecessor must be  $p$  or  $q$ . Otherwise both  $pz$  and  $qz$  are extra good edges for  $\{a, t\}$ , giving  $D_G(a, t) \geq 2$ . If its predecessor is  $p$  or  $q$ , it still contributes at least one. Thus  $z$  has at most one same-layer neighbor.

Choose a breadth-first tree by making  $p$  the predecessor of  $z$ . All non-tree edges are sibling edges, except  $qz$  and the possible same-layer edge incident with  $z$ . These exceptional edges lie entirely among proper descendants of  $w$ . As in Case 1, 2-connectivity forces  $w = a$ . Thus  $z \in L_2$ . The same subtree-separation argument rules out all later layers.

If  $z$  has no same-layer neighbor, every other predecessor class in  $L_2$  would be separated by its predecessor, so  $L_2 = \{z\}$ .

Otherwise let  $t$  be its sole same-layer neighbor, and exchange  $p, q$  if necessary so that  $p$  is the predecessor of  $t$ . All remaining vertices of  $L_2$  must have predecessor  $p$ , and their induced graph must be connected and contain  $t$ ; any other component would be separated by its predecessor. This is Type I.

Finally, if  $c_G(a) \leq 1$ , then (5) forces  $M = \emptyset$ , and (6) forces  $F = \emptyset$ . The argument in Case 1 then makes  $a$  universal. This proves (4).  $\square$

## 4. Consequences of the low-contribution structure

### Lemma 4.1 – A 2-connected deletion exists

Let  $G$  be 2-connected with no universal vertex. If some vertex has contribution at most three, then either  $G \cong C_5$ , or there are distinct vertices  $u, v$  such that

$$N[u] \subseteq N[v] \quad \text{and} \quad G - u \text{ is 2-connected.} \quad (7)$$

**Proof** Apply Lemma 3.1.

**Type I** If  $X = \emptyset$ , then  $z$  is simplicial, with adjacent neighbors  $p, q$ , so deleting  $z$  preserves 2-connectivity.

Suppose  $X \neq \emptyset$ , and let  $zt$  be its sole edge to  $z$ . Every vertex of  $X$  is dominated by  $p$ .

If  $X = \{t\}$ , then  $t$  is simplicial, with adjacent neighbors  $p, z$ , and can be deleted.

Otherwise choose  $u \in X \setminus \{t\}$  such that  $G[X] - u$  remains connected. Such a vertex exists by taking a leaf other than  $t$  in a spanning tree of  $G[X]$ .

The graph  $G - p$  is connected, and  $X$  is attached to its complement there only through  $tz$ . Deleting  $u$  preserves this connectivity. Thus  $G - \{u, p\}$  is connected. Since  $u$  is dominated by  $p$ , Lemma 2.1 shows that  $G - u$  is 2-connected.

**Type II** If  $|X| \geq 2$ , choose a non-cut vertex  $u \neq x$  of  $G[X]$ . It is dominated by  $p$ . In  $G - p$ , the set  $X$  is attached to its complement only by  $xy$ , and deleting  $u$  preserves connectivity. Lemma 2.1 again gives a 2-connected deletion. The case  $|Y| \geq 2$  is symmetric.

We may therefore assume  $X = \{x\}$  and  $Y = \{y\}$ .

If  $A = \{p, q\}$ , then:

- if  $pq \notin E(G)$ , the graph is  $C_5$ ;
- if  $pq \in E(G)$ , the vertex  $a$  is simplicial and can be deleted.

Finally suppose  $|A| \geq 3$ . The graph

$$R = G[A] + pq$$

is connected, because  $G - a$  is connected and the path  $p - x - y - q$  connects  $p$  to  $q$ .

Choose a spanning tree of  $R$  containing  $pq$ . Since this tree has at least three vertices, it has a leaf  $u \notin \{p, q\}$ . Thus  $R - u$  is connected. The vertex  $u$  is dominated by  $a$ , and replacing the added edge  $pq$  by  $p - x - y - q$  shows that  $G - \{a, u\}$  is connected. Lemma 2.1 proves that  $G - u$  is 2-connected.  $\square$

The next lemma prevents low-increment simplicial deletions in a counterexample without a universal vertex.

**Lemma 4.2 – Low-contribution simplicial vertices**

Let  $G$  be 2-connected and have no universal vertex. If  $a$  is simplicial and  $c_G(a) \leq 3$ , then

$$\eta(G) \geq 2|V(G)|. \quad (8)$$

**Proof** In Lemma 3.1,  $A = N(a)$  is now a clique.

Type I is impossible: the vertex  $p$  is adjacent to  $a$ , all of  $A$ ,  $z$ , and all of  $X$ , so it is universal.

Thus Type II holds. Put

$$K = \{a\} \cup A.$$

The vertices  $p, x, y, q$  induce the cycle

$$p - x - y - q - p.$$

It is isometric, so the sum of  $D_G$  over pairs within these four vertices is at least

$$\eta(C_4) = 8. \quad (9)$$

We show that every other vertex supplies at least two additional units, using only pairs with these four vertices.

**Vertices in  $K \setminus \{p, q\}$**  For  $b \in K \setminus \{p, q\}$ , the path  $b - p - x$  is shortest, and  $qy$  is an additional good edge for  $\{b, x\}$ . Hence

$$D_G(b, x) \geq 1.$$

Symmetrically,

$$D_G(b, y) \geq 1.$$

**Vertices in  $X \setminus \{x\}$**  Let  $b \in X \setminus \{x\}$ .

If  $bx \in E(G)$ , then:

- $b - p - q$  is shortest and  $xy$  is an additional good edge for  $\{b, q\}$ ;
- $b - x - y$  is shortest and  $pq$  is an additional good edge for  $\{b, y\}$ .

Thus  $D_G(b, q) + D_G(b, y) \geq 2$ .

If  $bx \notin E(G)$ , then  $d(b, y) = 3$ , and both

$$b - p - q - y, \quad b - p - x - y$$

are shortest paths. Their union has five edges, so

$$D_G(b, y) \geq 2.$$

The argument for  $Y \setminus \{y\}$  is symmetric.

These counted pairs are distinct: each has one endpoint outside the four-cycle and one inside it. All uncounted contributions are nonnegative. Therefore

$$\eta(G) \geq 8 + 2(|V(G)| - 4) = 2|V(G)|.$$

□

Combining the preceding lemmas gives the reduction needed for induction.

**Proposition 4.3 – Reduction with increment at least four**

Let  $G$  be 2-connected, with no universal vertex, and suppose

$$\eta(G) < 2|V(G)|.$$

Then either  $G \cong C_5$ , or there is a vertex  $u$  dominated by another vertex such that  $H = G - u$  is 2-connected and

$$\boxed{\eta(G) - \eta(H) \geq 4.} \quad (10)$$

**Proof** By (2), some vertex has contribution at most three. Unless  $G = C_5$ , Lemma 4.1 supplies the required dominated vertex  $u$  with a 2-connected deletion.

If  $u$  is not simplicial, then Lemma 3.1 gives  $c_G(u) \geq 2$ , and Lemma 2.1 gives  $q_G(u) \geq 2$ . Hence the increment is at least four.

If  $u$  is simplicial, Lemma 4.2 and the hypothesis  $\eta(G) < 2|V(G)|$  force  $c_G(u) \geq 4$ . Equation (3) again gives the result.

□

## 5. Graphs with a universal vertex

This section verifies the relevant result from the supplied attempt.

For a graph  $Q$ , let  $p_3(Q), p_4(Q), c_4(Q)$  count induced copies of  $P_3, P_4, C_4$ , respectively.

**Lemma 5.1**

If  $\text{diam}(G) \leq 2$ , then

$$\eta(G) = 2(p_3(G) - \overline{m}(G)) + p_4(G) + 4c_4(G), \quad (11)$$

where  $\overline{m}(G)$  is the number of nonedges.

**Proof** For a nonedge  $ab$ , every common neighbor gives two good edges. Every other good edge is the middle edge of an induced  $P_4$  with endpoints  $a, b$ . Thus

$$D_G(a, b) = 2(|N(a) \cap N(b)| - 1) + \#\{\text{induced } P_4\text{'s with endpoints } a, b\}.$$

For an edge  $ab$ , every good edge other than  $ab$  is the edge opposite  $ab$  in an induced  $C_4$ . Summing over pairs counts each induced  $C_4$  four times. Summing common-neighbor counts over nonedges counts  $p_3(G)$ . This proves (11). □

In particular, for a cone  $G = K_1 \vee Q$ ,

$$\boxed{\eta(G) = 2p_3(Q) + p_4(Q) + 4c_4(Q).} \quad (12)$$

Indeed, the apex creates one additional induced  $P_3$  for each nonedge of  $Q$ , and belongs to no induced  $P_4$  or  $C_4$ .

**Lemma 5.2**

Let  $Q$  be a connected, noncomplete graph of order  $m$ . Unless  $Q$  is  $K_m - e$  or a clique  $K_{m-1}$  with one pendant vertex,

$$2p_3(Q) + p_4(Q) + 4c_4(Q) \geq 3m - 7. \quad (13)$$

For each of the two exceptional graphs, the left side equals  $2m - 4$ .

**Proof** Write

$$F(Q) = 2p_3(Q) + p_4(Q) + 4c_4(Q).$$

Choose a maximum clique  $C$ , of order  $c \geq 2$ . Let  $L_i$  now denote distance layers from  $C$ , and put

$$\ell = |L_1|, \quad r = \sum_{i \geq 2} |L_i|.$$

Thus  $m = c + \ell + r$ , and  $\ell \geq 1$ .

For  $x \in L_1$ , set  $a_x = |N(x) \cap C|$ . Maximality of  $C$  gives

$$1 \leq a_x \leq c - 1.$$

The vertex  $x$ , one neighbor in  $C$ , and one nonneighbor in  $C$  give

$$a_x(c - a_x) \geq c - 1$$

induced  $P_3$ 's.

Every vertex in  $L_i$ ,  $i \geq 2$ , supplies another induced  $P_3$ , using the first three vertices of a shortest path to  $C$ . These copies are distinct, by their unique vertex of maximum layer. Consequently,

$$p_3(Q) \geq (c - 1)\ell + r. \quad (14)$$

Likewise,

$$p_4(Q) \geq r. \quad (15)$$

For a vertex at distance at least three, take four vertices of a shortest path. For  $x \in L_2$ , take  $x - y - v$  with  $v \in C$ , and choose  $w \in C \setminus N(y)$ ; then  $x - y - v - w$  is induced. Again the maximum-layer vertex distinguishes the copies.

**Case 1:**  $c \geq 3$  and  $\ell \geq 2$  Equations (14)–(15) give

$$F(Q) \geq 2(c - 1)\ell + 3r.$$

The difference from  $3m - 7$  is

$$(2c - 5)\ell - 3c + 7 \geq c - 3 \geq 0.$$

**Case 2:**  $c \geq 3$  and  $\ell = 1$  Write  $L_1 = \{x\}$ ,  $a = |N(x) \cap C|$ , and  $A = a(c - a) \geq c - 1$ .

If  $r = 0$ , then  $F(Q) = 2A$ . The cases  $a = 1$  and  $a = c - 1$  are the two exceptions. Otherwise  $2 \leq a \leq c - 2$ , and

$$F(Q) \geq 4(c - 2) \geq 3c - 4 = 3m - 7.$$

Suppose  $r > 0$ , and write

$$s = |L_2|, \quad t = \sum_{i \geq 3} |L_i|.$$

Every vertex of  $L_2$  is adjacent to  $x$ . It supplies  $A$  induced paths  $y - x - v - w$ , where  $v \in N(x) \cap C$  and  $w \in C \setminus N(x)$ . Hence

$$p_3(Q) \geq A + s + t, \quad p_4(Q) \geq As + t.$$

Therefore

$$F(Q) \geq 2A + (A + 2)s + 3t.$$

Subtracting  $3m - 7 = 3c + 3s + 3t - 4$ , and using  $s \geq 1$ , gives

$$2A + (A - 1)s - 3c + 4 \geq 2(c - 1) + (c - 2) - 3c + 4 = 0.$$

**Case 3:**  $c = 2$  Write  $C = \{u, v\}$ . The graph is triangle-free. Partition  $L_1$  into

$$A = N(u) \cap L_1, \quad B = N(v) \cap L_1,$$

with sizes  $\alpha, \beta$ , so  $\ell = \alpha + \beta$ . Let  $e$  be the number of edges between  $A$  and  $B$ .

Besides the copies in (14), pairs in  $A$  and in  $B$  give

$$\binom{\alpha}{2} + \binom{\beta}{2}$$

additional induced  $P_3$ 's. A nonedge between  $A, B$  gives an induced  $P_4$  through  $u, v$ , and an edge between them gives an induced  $C_4$ . These copies are disjoint from those counted in (15). Thus

$$\begin{aligned} F(Q) &\geq 2 \left( \ell + r + \binom{\alpha}{2} + \binom{\beta}{2} \right) + r + (\alpha\beta - e) + 4e \\ &= 2\ell + 3r + \alpha(\alpha - 1) + \beta(\beta - 1) + \alpha\beta + 3e. \end{aligned}$$

For nonnegative integers  $\alpha, \beta$  with  $\alpha + \beta \geq 1$ ,

$$\alpha(\alpha - 1) + \beta(\beta - 1) + \alpha\beta \geq \alpha + \beta - 1.$$

If both are positive, this follows by bounding the first two terms by  $\alpha - 1, \beta - 1$ , and  $\alpha\beta$  by 1; if one is zero it is immediate. Hence

$$F(Q) \geq 3\ell + 3r - 1 = 3m - 7.$$

Finally, each exceptional graph has  $m - 2$  induced  $P_3$ 's and no induced  $P_4$  or  $C_4$ , giving  $F(Q) = 2m - 4$ .  $\square$

### Proposition 5.3

Let  $G$  be 2-connected of order  $n$ , with a universal vertex. If

$$G \not\cong K_n, K_n^2, K_n^{n-2},$$

then

$$\boxed{\eta(G) \geq 3n - 10.} \quad (16)$$

**Proof** Write  $G = K_1 \vee Q$ . The graph  $Q$  is connected, since deleting the universal vertex preserves connectivity.

If  $Q$  is complete,  $G = K_n$ . The two exceptions of Lemma 5.2 give, respectively,  $K_n^{n-2}$  and  $K_n^2$ . Otherwise (12)–(13), with  $m = n - 1$ , give

$$\eta(G) \geq 3(n - 1) - 7 = 3n - 10.$$

$\square$

## 6. Proof of the theorem

First, formula (11) gives

$$\boxed{\eta(K_m^t) = 2(m - 1 - t)(t - 1).} \quad (17)$$

Indeed, there are  $m - 1 - t$  nonedges, each with  $t$  common neighbors, and no induced  $P_4$  or  $C_4$ . In particular,

$$\eta(K_m^2) = \eta(K_m^{m-2}) = 2m - 6. \quad (18)$$

Put

$$f(n) = \min\{2n, 3n - 10\}.$$

We prove (1) by induction on  $n$ .

For  $n = 4$ , the only admissible graph is  $C_4$ , and  $\eta(C_4) = 8 \geq f(4)$ .

Now let  $n \geq 5$ , and suppose the assertion holds at smaller orders.

If  $G$  has a universal vertex, Proposition 5.3 gives the result. We may therefore assume that  $G$  has no universal vertex.

If  $\eta(G) \geq 2n$ , there is nothing to prove. If  $G = C_5$ , then

$$\eta(G) = 20 - 15 = 5 = f(5).$$

Otherwise Proposition 4.3 supplies a dominated vertex  $u$  such that

$$H = G - u$$

is 2-connected and

$$\eta(G) \geq \eta(H) + 4. \quad (19)$$

Write  $m = n - 1$ .

**Case 1:  $H$  is not exceptional**

By induction,

$$\eta(G) \geq f(n - 1) + 4.$$

But

$$f(n - 1) + 4 = \min\{2n + 2, 3n - 9\} \geq \min\{2n, 3n - 10\} = f(n).$$

So this case is finished.

It remains to check every possible exceptional predecessor.

**Case 2:  $H \cong K_m$**

Every neighbor of  $u$  is universal in  $G$ , contrary to our assumption that  $G$  has no universal vertex.

**Case 3:  $H \cong K_m^{m-2} = K_m - e$**

Let  $x, y$  be the endpoints of its missing edge. Every vertex of  $H \setminus \{x, y\}$  is universal in  $H$ . Since  $G$  has no universal vertex, none is adjacent to  $u$ .

Because  $G$  is 2-connected,  $d_G(u) \geq 2$ , so

$$N_G(u) = \{x, y\}.$$

But neither  $x$  nor  $y$  can dominate  $u$ , since  $xy \notin E(G)$ . This contradicts the choice of  $u$ .

**Case 4:  $H \cong K_m^2$**

The case  $m = 4$  is already included in Case 3. Otherwise write  $H$  as a clique  $B$  of order  $m - 1$  and an additional vertex  $z$  with

$$N_H(z) = \{p, q\} \subseteq B.$$

Put

$$R = B \setminus \{p, q\}.$$

The vertices  $p, q$  are universal in  $H$ . Thus  $u$  is adjacent to neither of them.

We claim that  $u$  is not adjacent to  $z$ . Otherwise, since  $d_G(u) \geq 2$ , it also has a neighbor in  $R$ . A vertex dominating  $u$  would have to be a neighbor of  $u$ ; but  $z$  is adjacent to no vertex of  $R$ , and no vertex of  $R$  is adjacent to  $z$ . No possible dominator exists.

Consequently,

$$N_G(u) \subseteq R.$$

Let  $t = |N_G(u)| \geq 2$ . The vertex  $u$  is simplicial.

Each of  $p, q$  has all  $t$  vertices of  $N(u)$  as common neighbors with  $u$ , so

$$D_G(u, p) \geq 2(t-1), \quad D_G(u, q) \geq 2(t-1). \quad (20)$$

Also  $d_G(u, z) = 3$ . For every  $r \in N(u)$ , the two paths

$$u - r - p - z, \quad u - r - q - z$$

are shortest. Their union over all  $r$  has  $3t + 2$  edges, all good for  $\{u, z\}$ . Therefore

$$D_G(u, z) \geq 3t - 1. \quad (21)$$

Combining (20)–(21),

$$c_G(u) \geq 7t - 5 \geq 9.$$

By Lemma 2.1 and (18),

$$\eta(G) \geq (2m - 6) + 9 = 2n + 1.$$

This contradicts the case assumption  $\eta(G) < 2n$ .

All exceptional predecessors have now been covered, completing the induction and proving (1).  $\square$

Since  $3n - 10 \geq 2n$  exactly when  $n \geq 10$ , the conjecture follows.

## 7. Sharpness and scope

The stronger bound is sharp for every  $n \geq 5$ .

**The  $3n - 10$  branch** For

$$G = K_1 \vee P_{n-1},$$

equation (12) gives

$$\eta(G) = 2(n-3) + (n-4) = 3n - 10.$$

These graphs are admissible for  $n \geq 5$ .

**The  $2n$  branch** Start with the cycle  $p - x - y - q - p$ , and enlarge its edge  $pq$  to a clique by adding  $n - 4$  vertices, each adjacent to every vertex already in that clique and to neither  $x$  nor  $y$ .

Equivalently, take a clique of order  $n - 2$ , an additional edge  $xy$ , and exactly two cross edges forming a matching.

Each added vertex is simplicial and has contribution exactly two: its only nonzero contributions are one with  $x$  and one with  $y$ . Lemma 2.1 therefore gives

$$\eta(G) = \eta(C_4) + 2(n-4) = 2n.$$

Thus the transition at  $n = 10$  is genuine.

No computational enumeration or unproved external assertion is used. In particular, the previously untreated graphs without universal vertices are covered by Proposition 4.3, and the induction explicitly handles all three exceptional predecessor families.